

CS-663

Digital Image Processing (DIP)

Image Segmentation

Suyash P. Awate

Pseudocode

- en.wikipedia.org/wiki/Pseudocode#Mathematical_style_pseudocode

- Mathematical notation, typically from matrix and set theory,
- Control structures of a conventional programming language,
- Natural language descriptions

- Examples

$$\text{Return } \sum_{k \in S} x_k$$

algorithm ford-fulkerson **is**

input: Graph G with flow capacity c ,
source node s ,
sink node t

output: Flow f such that f is maximal from s to t

(Note that $f_{(u,v)}$ is the flow from node u to node v , and $c_{(u,v)}$ is the flow capacity from node u to node v)

for each edge (u, v) **in** G_E **do**

$f_{(u, v)} \leftarrow 0$

$f_{(v, u)} \leftarrow 0$

while there exists a path p from s to t **in** the residual network G_f **do**

let c_f be the flow capacity of the residual network G_f

$c_f(p) \leftarrow \min\{c_f(u, v) \mid (u, v) \text{ in } p\}$

for each edge (u, v) **in** p **do**

$f_{(u, v)} \leftarrow f_{(u, v)} + c_f(p)$

$f_{(v, u)} \leftarrow -f_{(u, v)}$

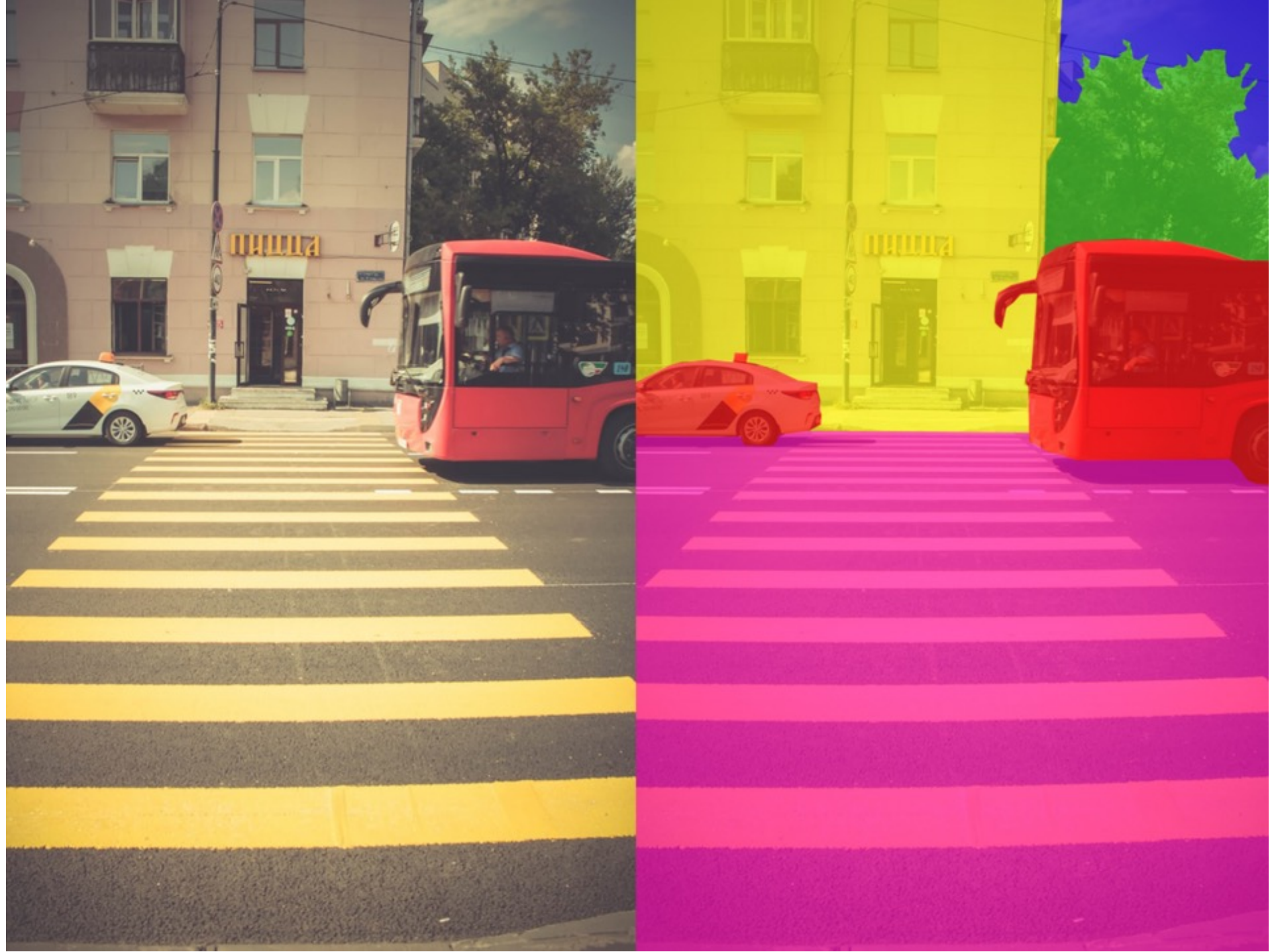
return f

Segmentation

- Partitioning an image into regions or segments
 - Per-pixel labeling task, i.e., task of assigning a label to each pixel
 - Label = categorical variable, e.g., {A, B, C}, {@, #, !}, etc.
 - All pixels with same label = one segment
 - Typically, 1 segment = 1 (or a few) connected regions
- Why ?
 - Simplification of information content
 - Towards higher-level understanding/interpretation
- Criteria
 - Small variation in intensity/texture within segment
 - Large variation in intensity/texture between segments
 - Similar to Otsu thresholding

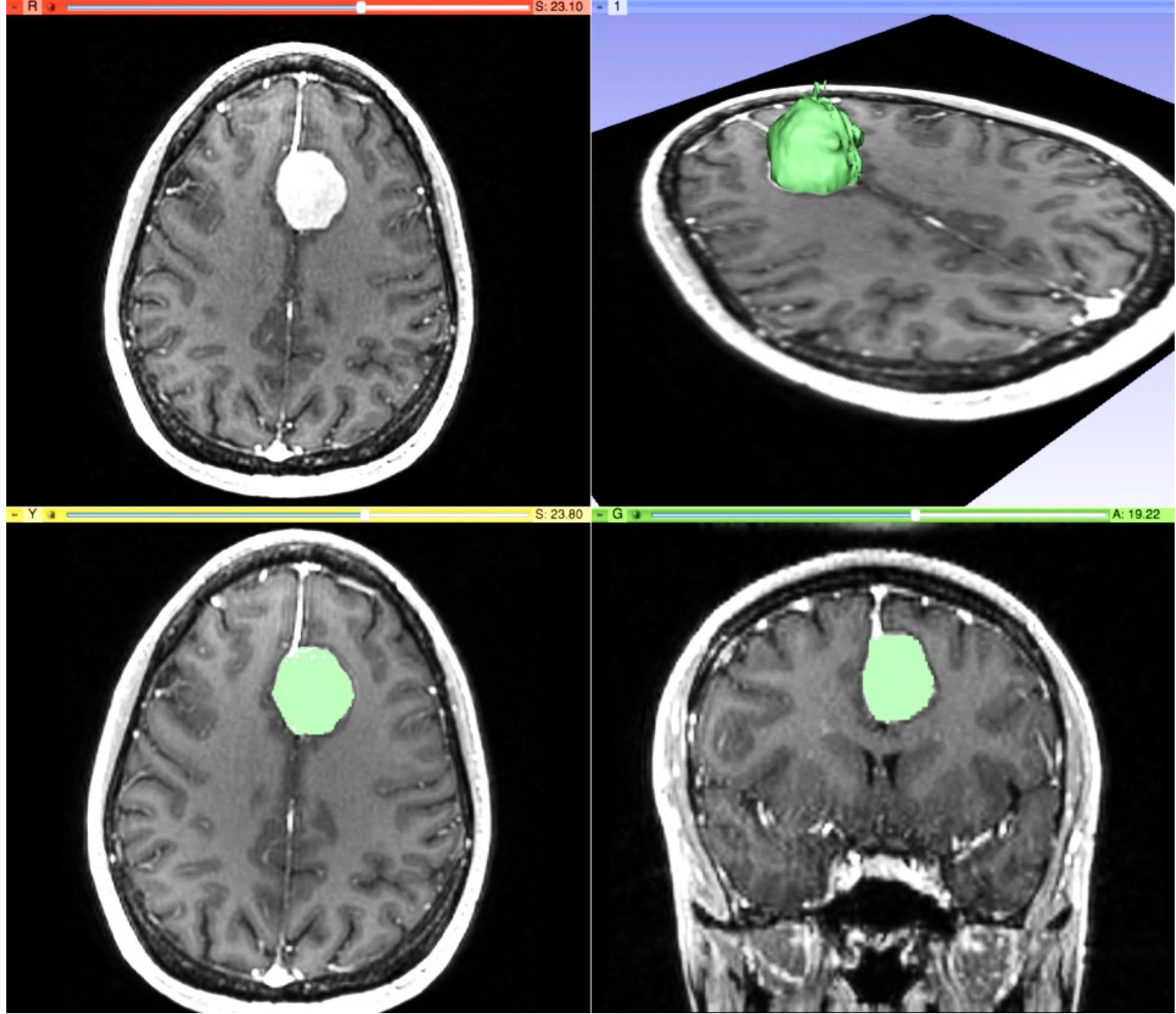
Segmentation

- Example



Segmentation

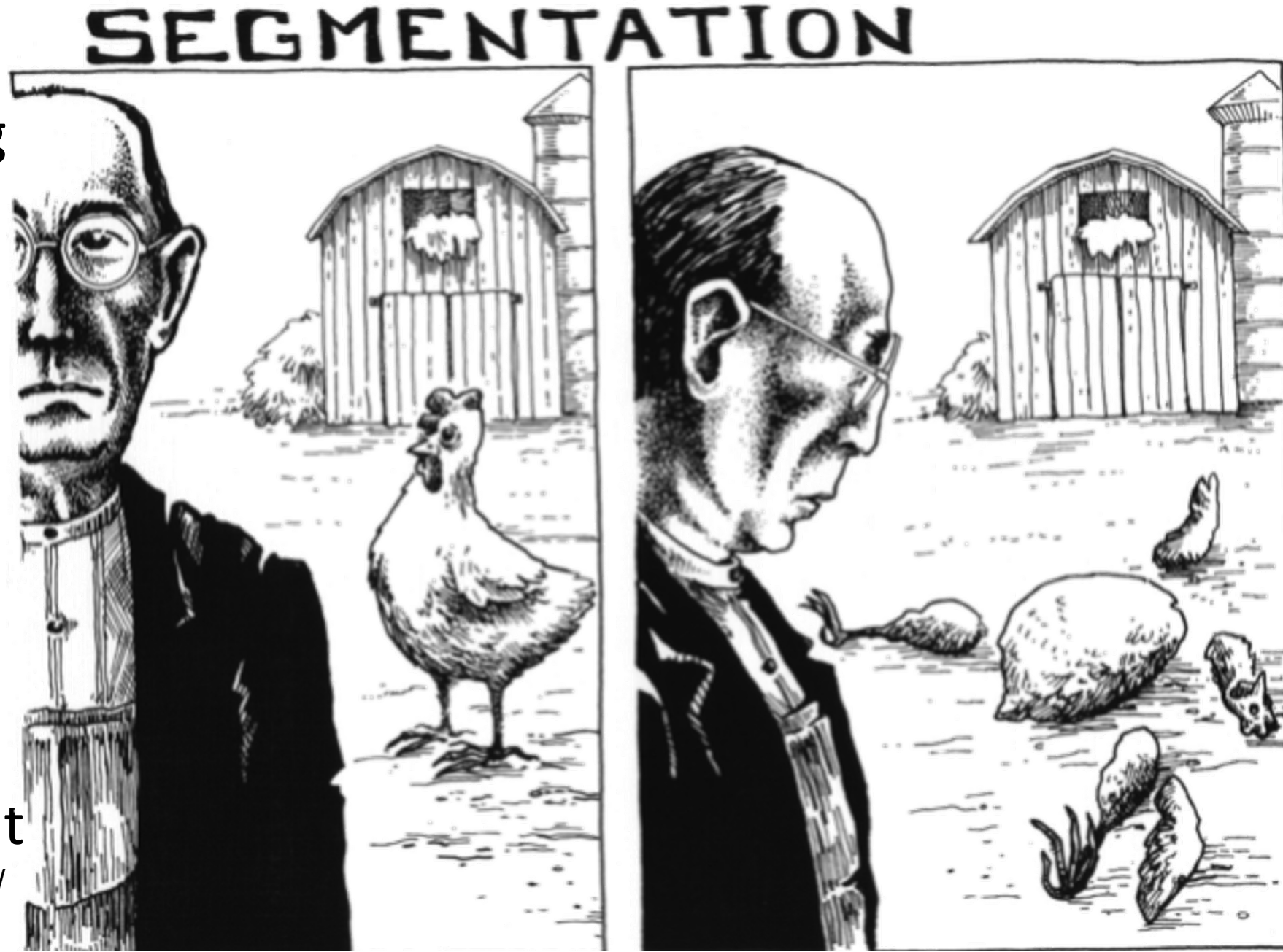
- Top left: Original Image with tumor
- Bottom: 2D views (left: axial, right: coronal)
- Top right: 3D visualization of segmented tumor



Segmentation

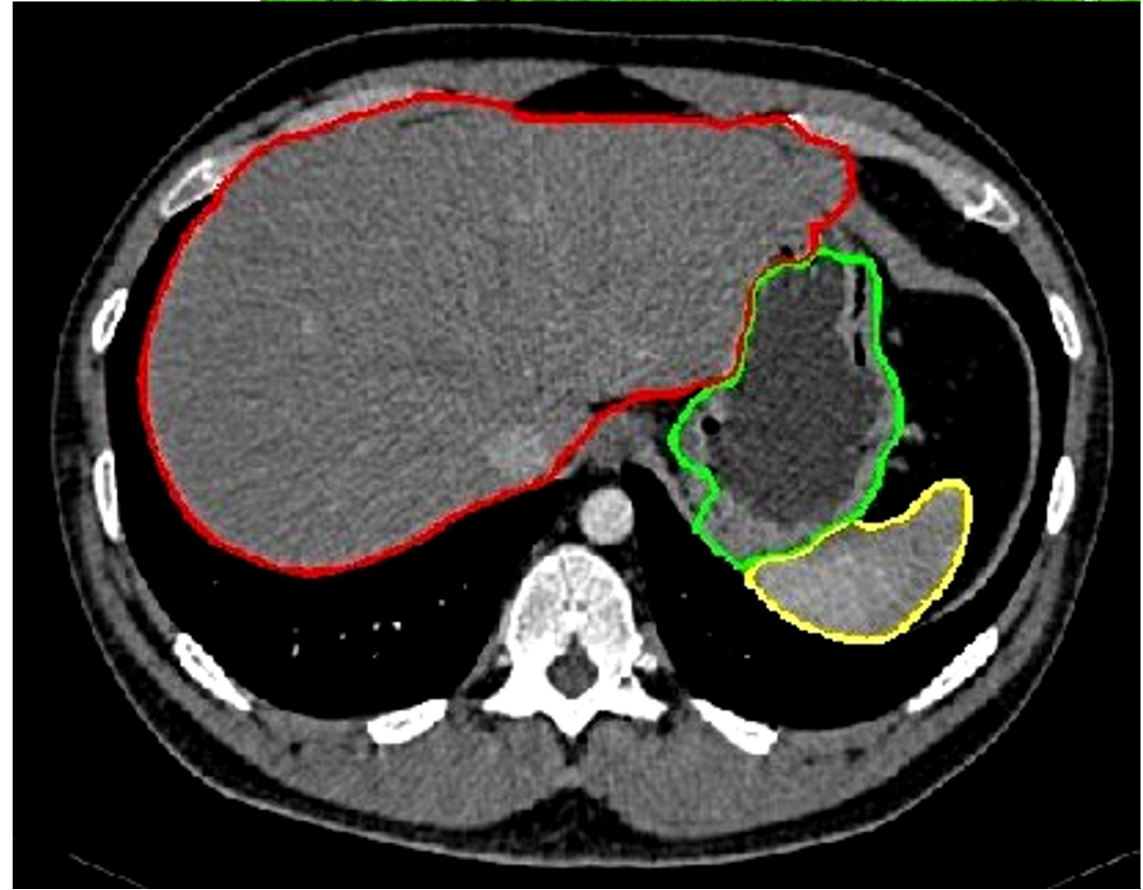
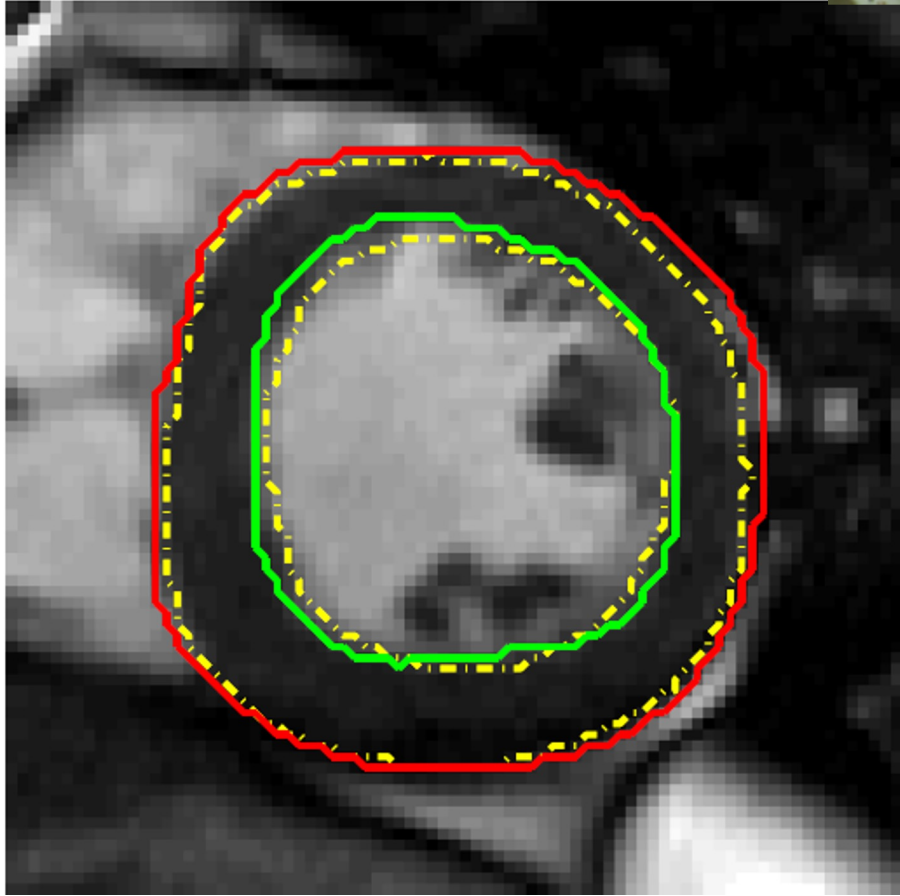
- Human perception
- Instead of perceiving indivisible objects, we perceive objects in terms of their labeled or categorized parts
- We **break** objects **into parts** that we have learned are relevant or important

cognitrn.psych.indiana.edu/rgoldsto/projects/segmentation.html



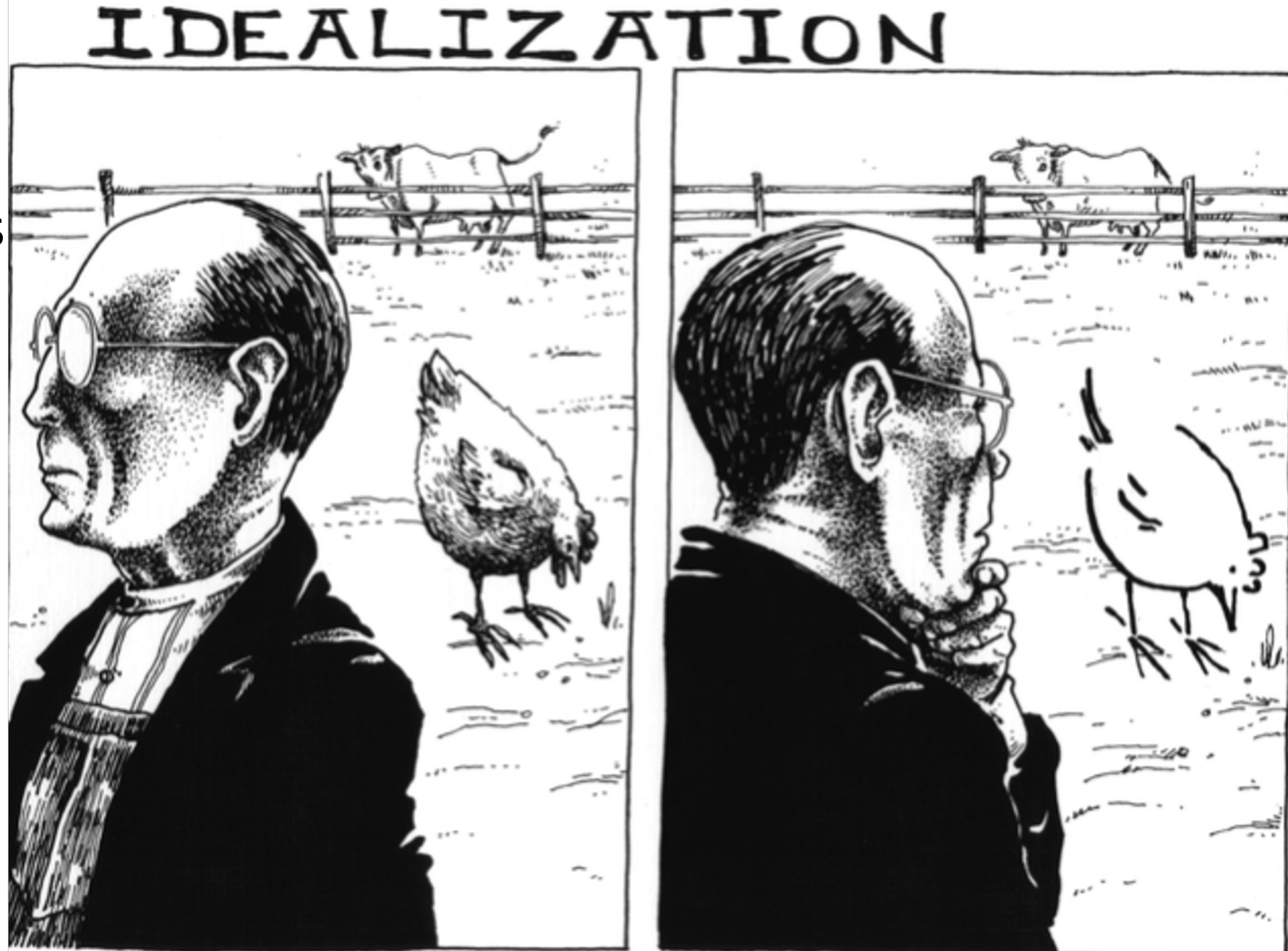
Segmentation

- **Delineation:**
Process of determining **outline** of an object in image



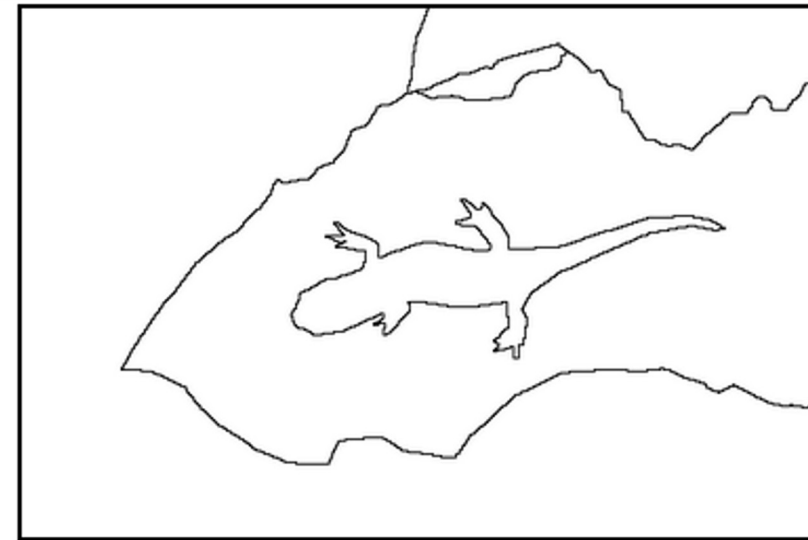
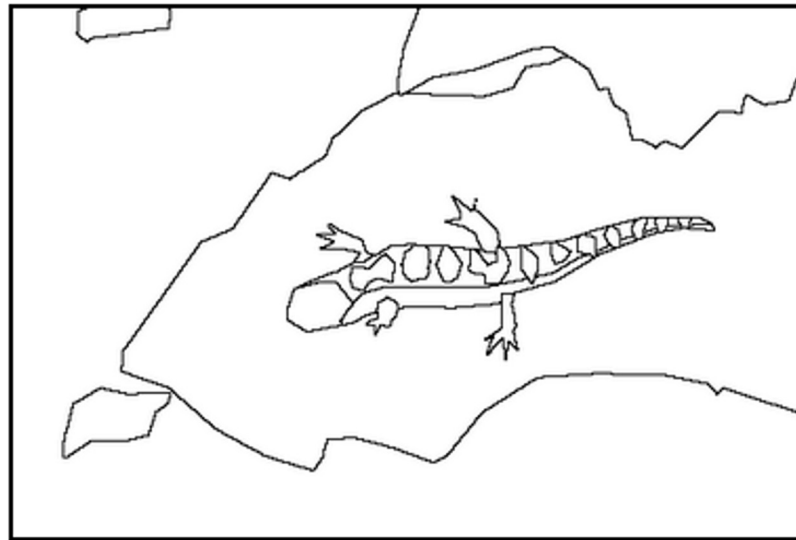
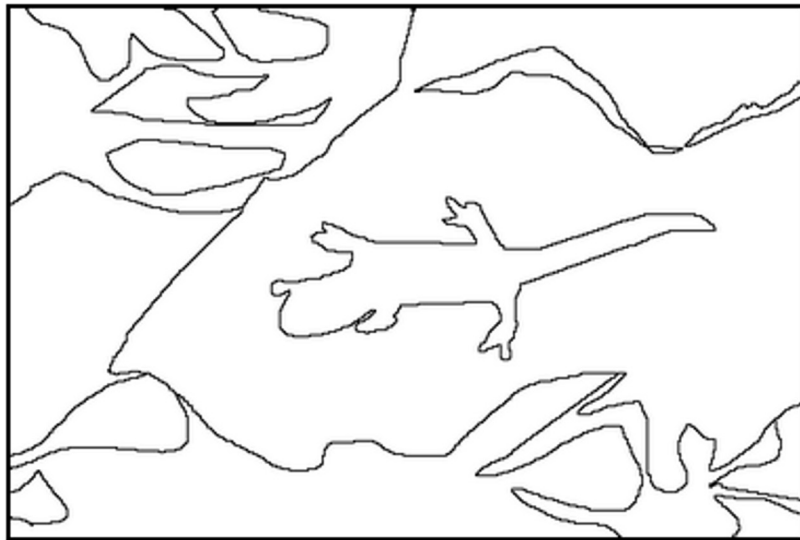
Segmentation

- Human perception (delineation)
- We simplify objects in order to capture basic essence of underlying concept



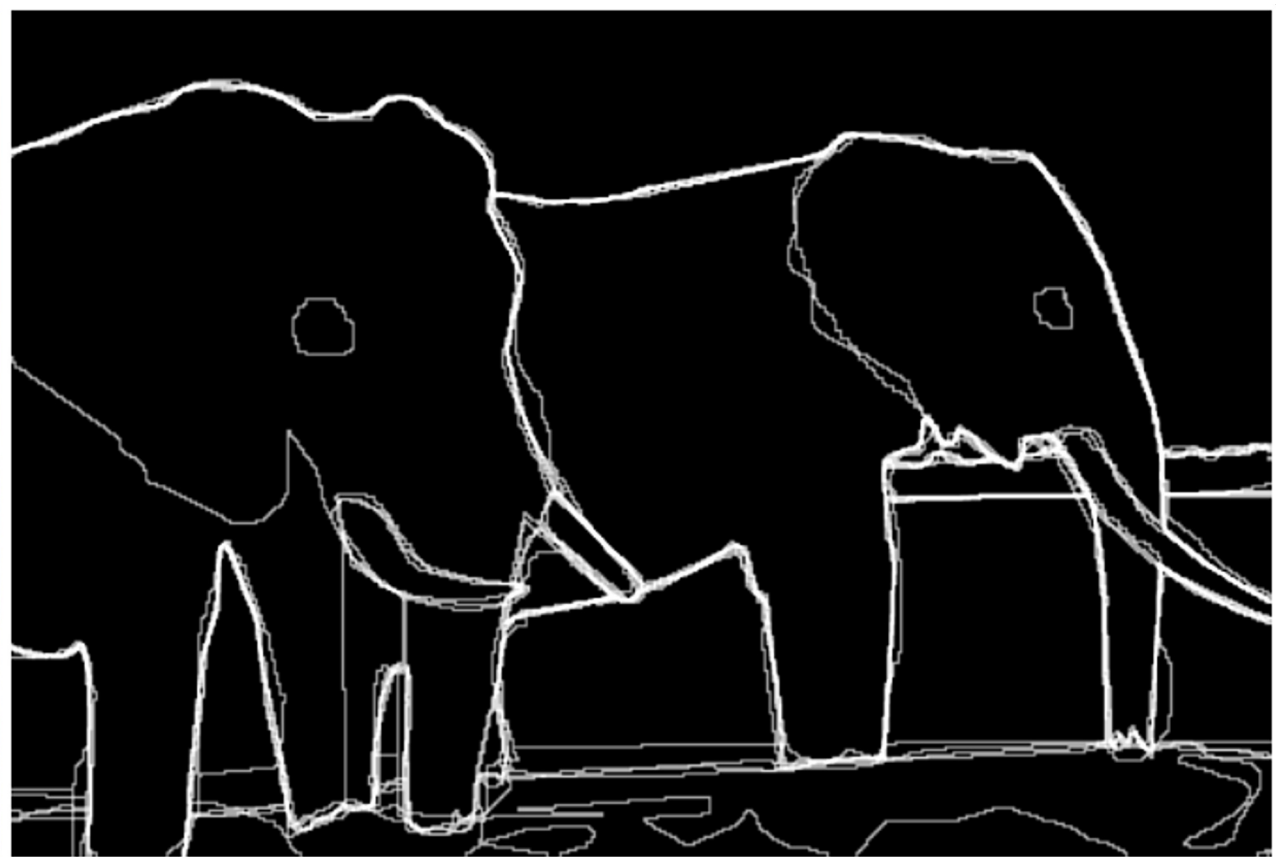
Segmentation

- Segmentation and delineation, both, can be subjective
- Different experts can give different answers



Segmentation

- Segmentation and delineation, both, can be subjective
- Different experts can give different answers



The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition

KEINOSUKE FUKUNAGA, SENIOR MEMBER, IEEE, AND LARRY D. HOSTETLER, MEMBER, IEEE

Abstract—Nonparametric density gradient estimation using a generalized kernel approach is investigated. Conditions on the kernel functions are derived to guarantee asymptotic unbiasedness, consistency, and uniform consistency of the estimates. The results are generalized to obtain a simple mean-shift estimate that can be extended in a k -nearest-neighbor approach. Applications of gradient estimation to pattern recognition are presented using clustering and intrinsic dimensionality problems, with the ultimate goal of providing further understanding of these problems in terms of density gradients.

Mean Shift: A Robust Approach Toward Feature Space Analysis

Dorin Comaniciu, *Member, IEEE*, and Peter Meer, *Senior Member, IEEE*

Abstract—A general nonparametric technique is proposed for the analysis of a complex multimodal feature space and to delineate arbitrarily shaped clusters in it. The basic computational module of the technique is an old pattern recognition procedure, the mean shift. We prove for discrete data the convergence of a recursive mean shift procedure to the nearest stationary point of the underlying density function and, thus, its utility in detecting the modes of the density. The relation of the mean shift procedure to the Nadaraya-Watson estimator from kernel regression and the robust M-estimators of location is also established. Algorithms for two low-level vision tasks, discontinuity preserving smoothing and image segmentation, are described as applications. In these algorithms, the only user set parameter is the resolution of the analysis and either gray level or color images are accepted as input. Extensive experimental results illustrate their excellent performance.

Index Terms—Mean shift, clustering, image segmentation, image smoothing, feature space, low-level vision.

Density Estimation

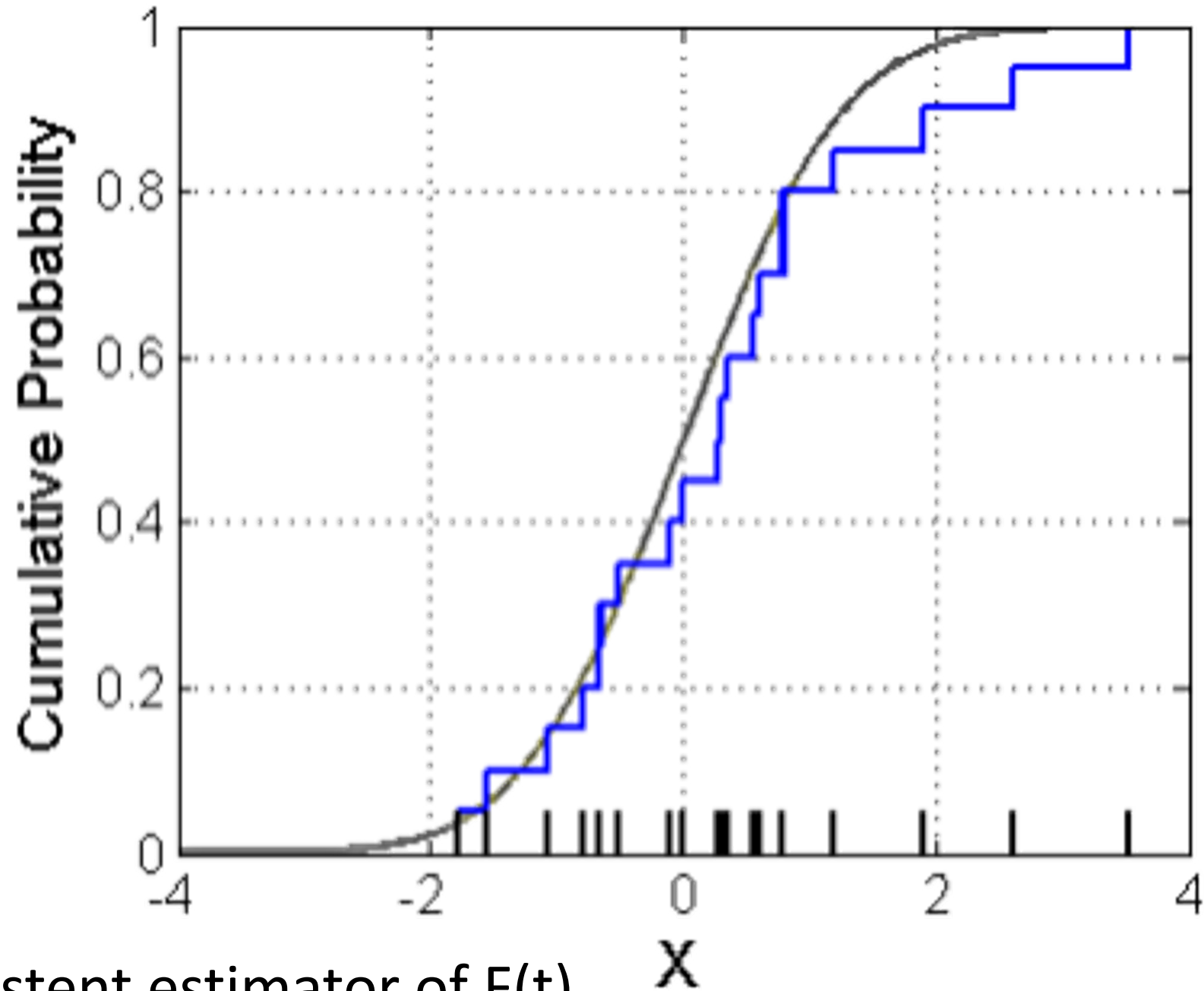
- Goal: Estimate **probability density function (PDF) $P(X)$** given observations $\{x_i \text{ in } \mathbb{R}^d : i = 1, \dots, n\}$ drawn from $P(X)$, where analytical/parametric form of $P(x)$ is unknown
- For PDF $P(X)$, let **cumulative distribution function (CDF) be $F(X)$**
 - Notation: Let $F(t)$ denote $F(X=t)$
- Let random variables $\{X_i \sim P(X) : i = 1, \dots, n\}$ be independent
- **Empirical distribution function (EDF).**
EDF(t) estimates CDF(t) by counting fraction of n draws $\leq t$

$$\hat{F}_n(t) = \frac{\text{number of elements in the sample } \leq t}{n} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{X_i \leq t}$$

- Indicators $\{\mathbf{1}_{X_i \leq t} : \text{for all } i\}$ are i.i.d. Bernoulli with parameter $F(t)$

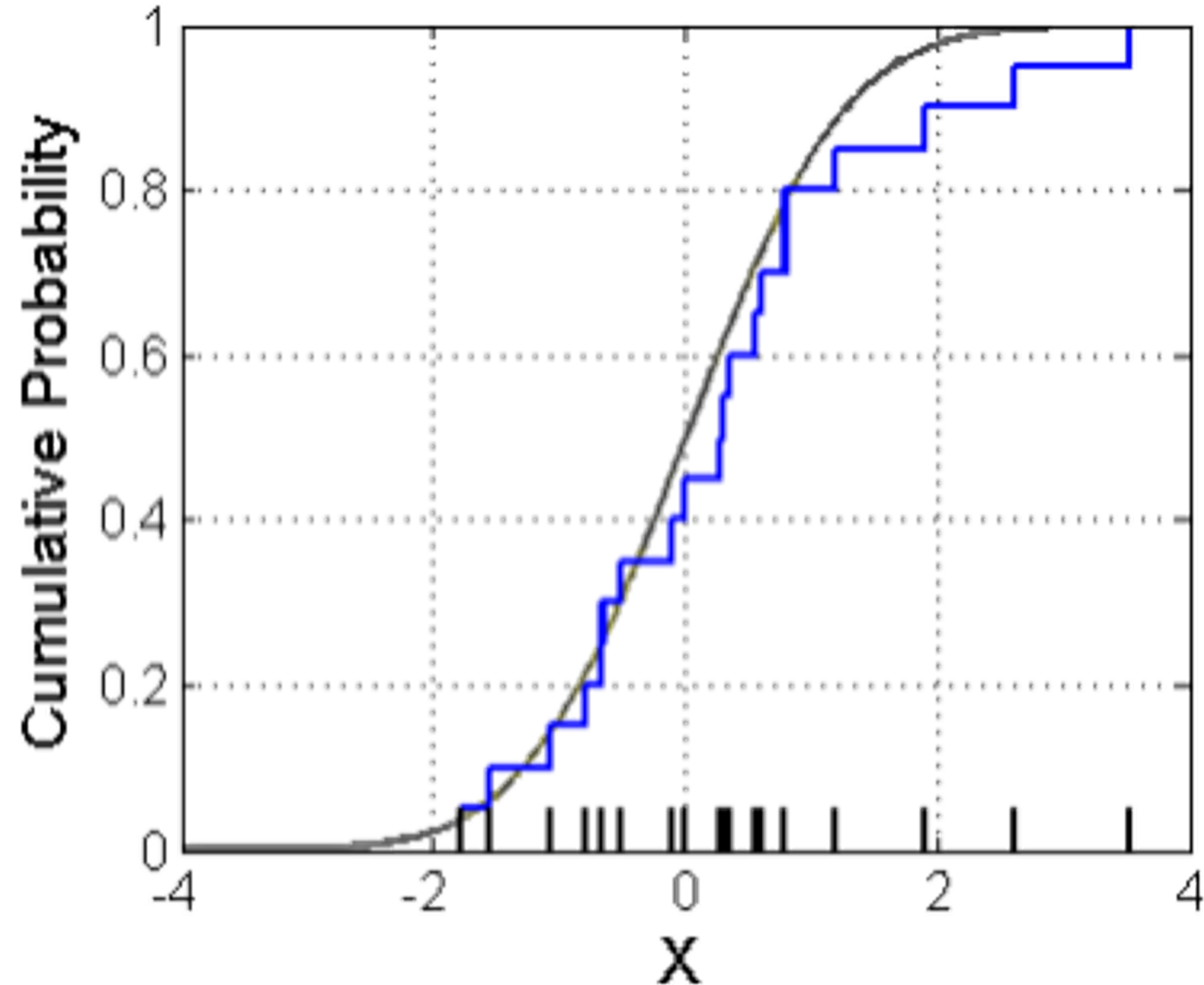
Density Estimation

- $n\hat{F}_n(t)$ =
sum of i.i.d.
Bernoulli variables
- $n\hat{F}_n(t)$ is Binomial;
mean $nF(t)$;
variance $nF(t)(1-F(t))$
- As $n \rightarrow \infty$,
 $\hat{F}_n(t)$ has
mean = $nF(t) / n = F(t)$;
variance =
 $nF(t)(1-F(t)) / n^2 \rightarrow 0$
- So $\hat{F}_n(t)$ is unbiased consistent estimator of $F(t)$



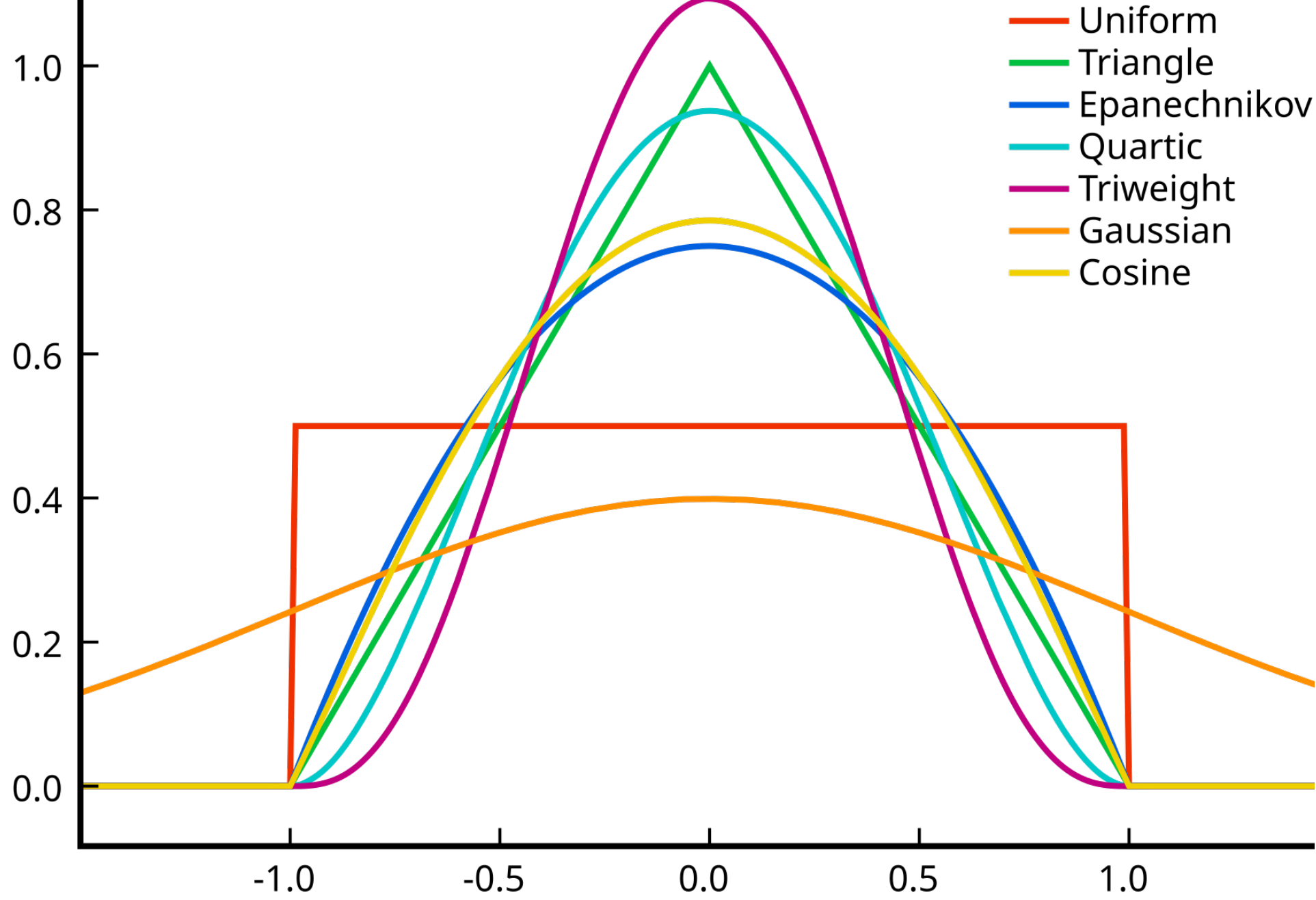
Density Estimation

- EDF is piecewise-constant
- Consider derivative of EDF to approximate PDF
- Derivative = sum of shifted impulses $\delta_{x_i}(x)$; non-smooth
- If we convolve derivative with **smooth** kernel function $k(x)$:
- $(\delta_{x_i} * k)(y)$
- $:= \int \delta(x - x_i)k(y - x)dx$
- $= \int \delta(z)k(y - x_i - z)dz = k(y - x_i) \rightarrow$ kernel $k(\cdot)$ shifted by x_i
- Then we get: mixture of kernels $\rightarrow$ kernel density estimation



Nonparametric/Kernel Density Estimation

- Common kernel functions



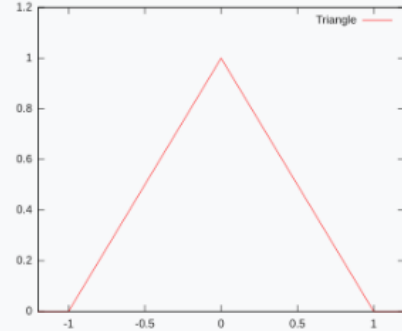
Mean-Shift Segm

- Finite-support kernels

Triangular

$$K(u) = (1 - |u|)$$

$$\text{Support: } |u| \leq 1$$

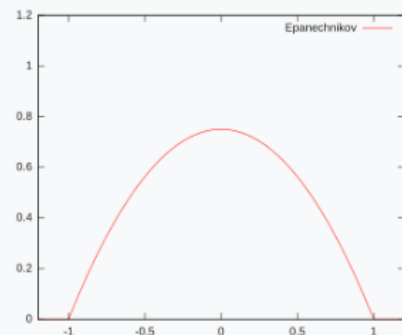


Epanechnikov

(parabolic)

$$K(u) = \frac{3}{4}(1 - u^2)$$

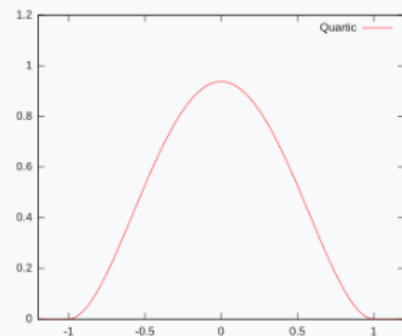
$$\text{Support: } |u| \leq 1$$



Quartic
(biweight)

$$K(u) = \frac{15}{16}(1 - u^2)^2$$

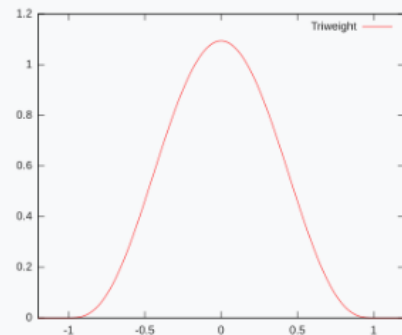
$$\text{Support: } |u| \leq 1$$



Triweight

$$K(u) = \frac{35}{32}(1 - u^2)^3$$

$$\text{Support: } |u| \leq 1$$

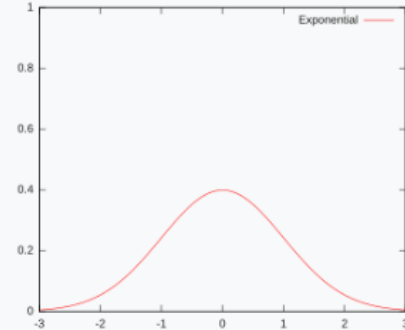


Mean-Shift Segm

- Kernel functions

Gaussian

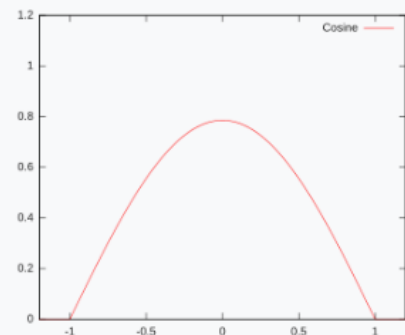
$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}$$



Cosine

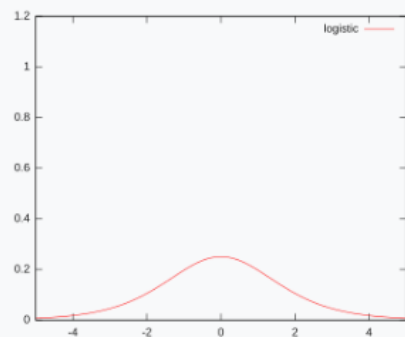
$$K(u) = \frac{\pi}{4} \cos\left(\frac{\pi}{2}u\right)$$

Support: $|u| \leq 1$



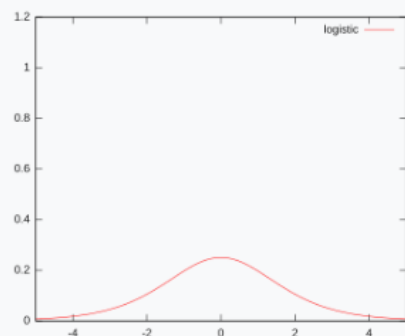
Logistic

$$K(u) = \frac{1}{e^u + 2 + e^{-u}}$$



**Sigmoid
function**

$$K(u) = \frac{2}{\pi} \frac{1}{e^u + e^{-u}}$$



Mean-Shift Segmentation

- Finite-support kernel:
[wiki/Truncated normal distribution](https://en.wikipedia.org/wiki/Truncated_normal_distribution)

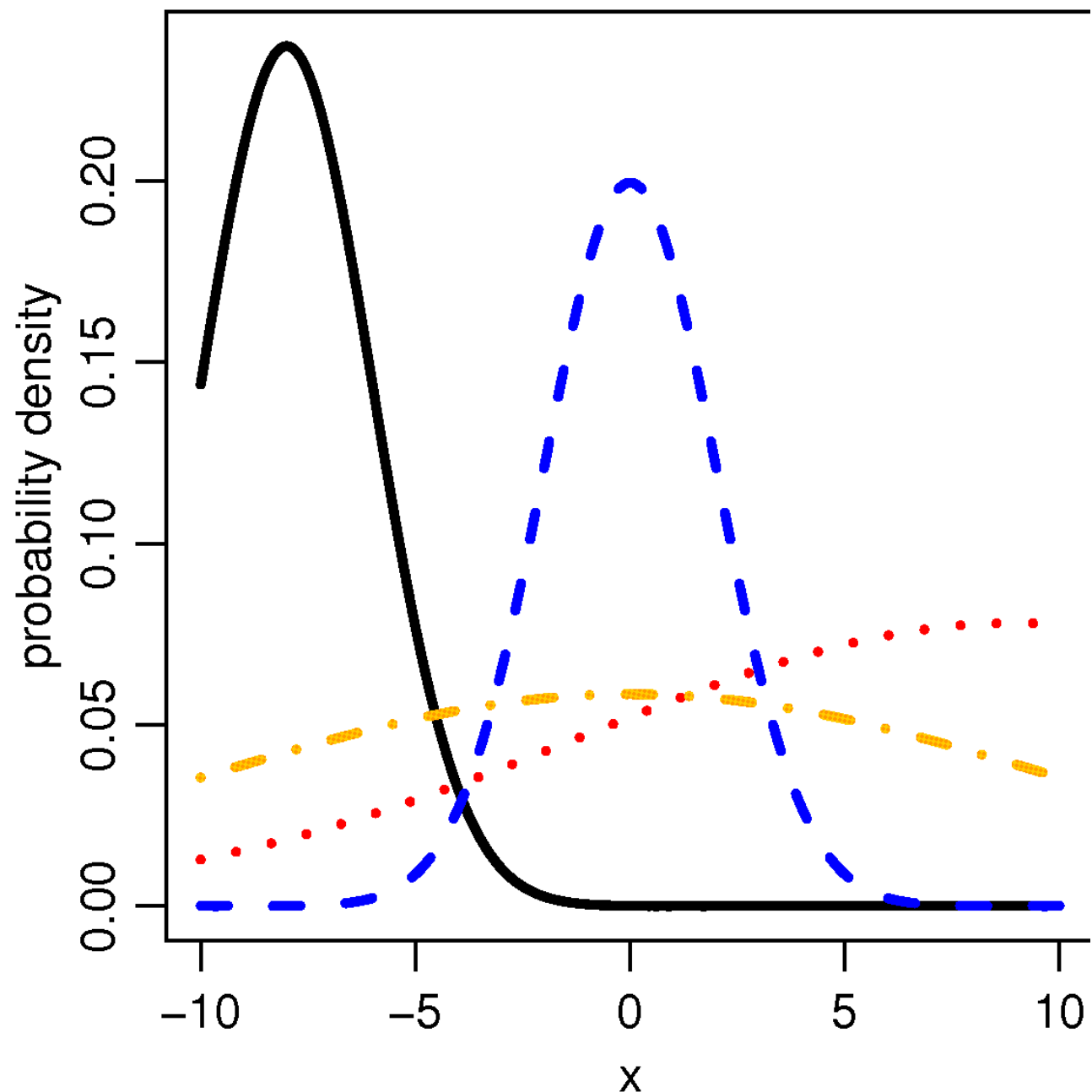
Its **probability density function**, f , for $a \leq x \leq b$, is

$$f(x; \mu, \sigma, a, b) = \frac{1}{\sigma} \frac{\varphi\left(\frac{x-\mu}{\sigma}\right)}{\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)}$$

and by $f = 0$ otherwise.

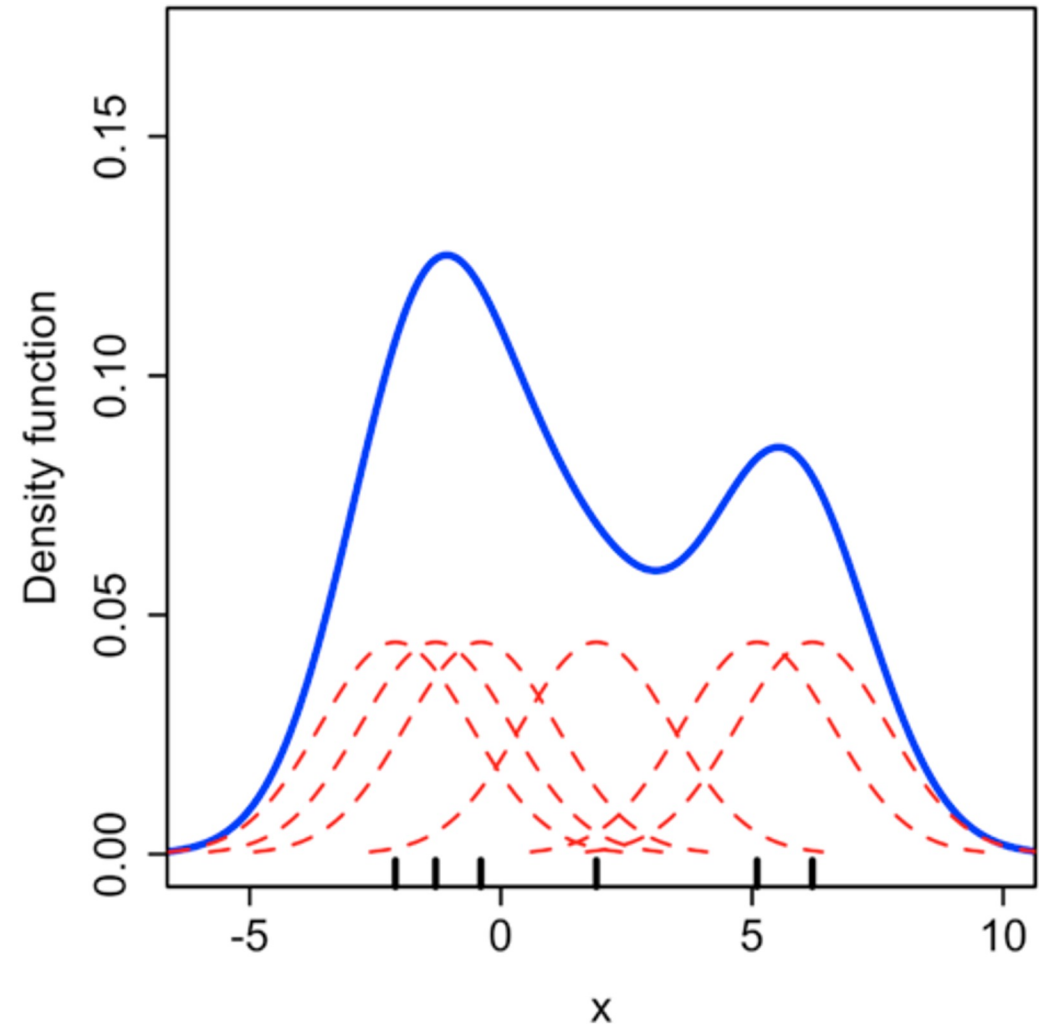
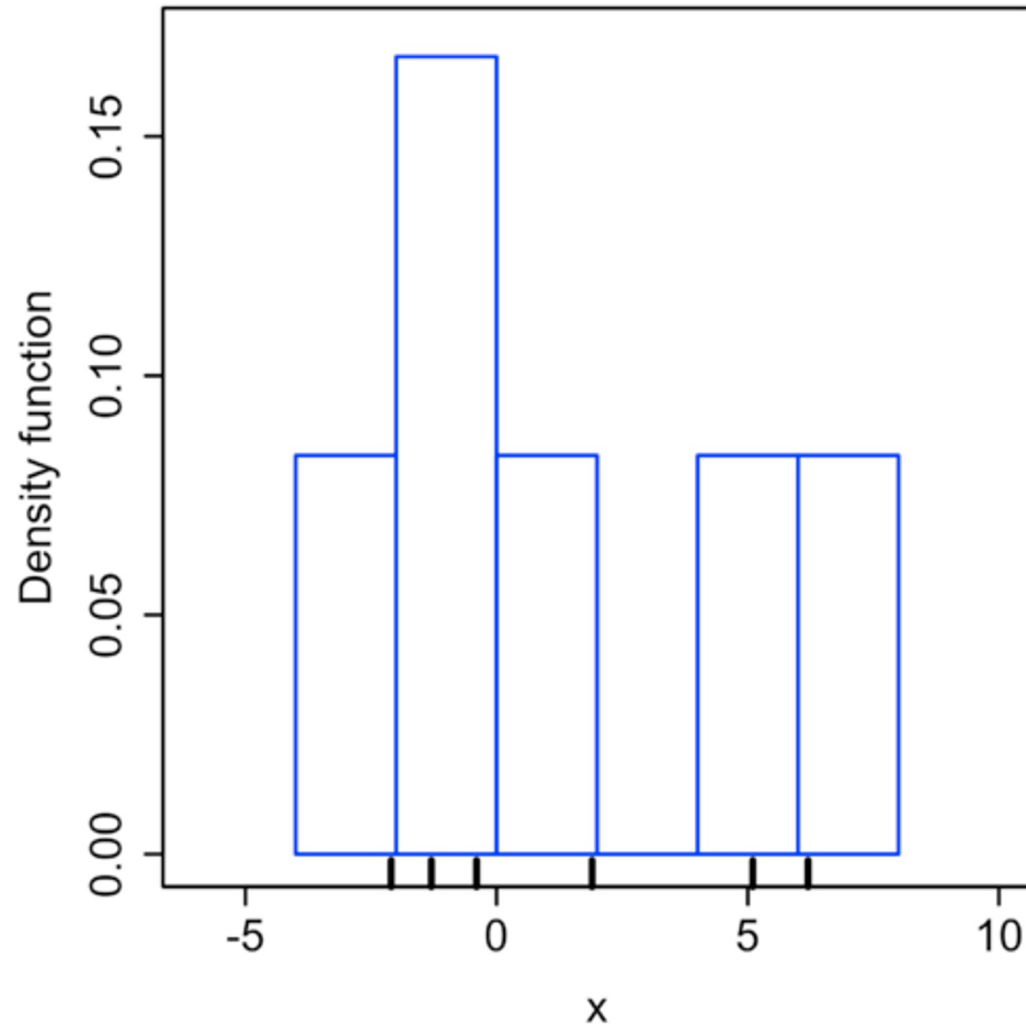
$$\varphi(\xi) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\xi^2\right)$$

$$\Phi(x) = \frac{1}{2} (1 + \operatorname{erf}(x/\sqrt{2}))$$



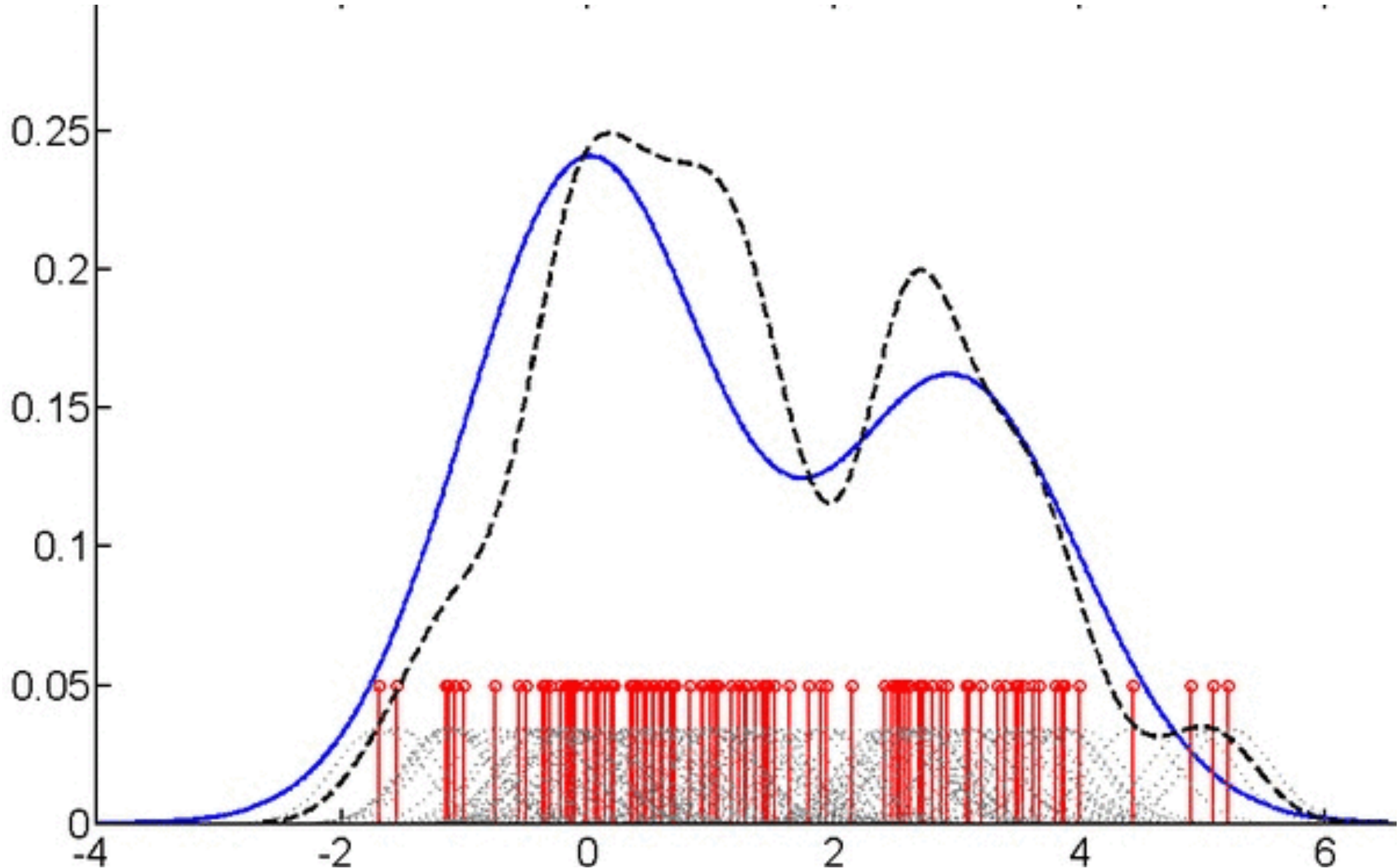
Nonparametric/Kernel Density Estimation

- Left: histogram
- Right: kernel density estimate (with Gaussian kernel)



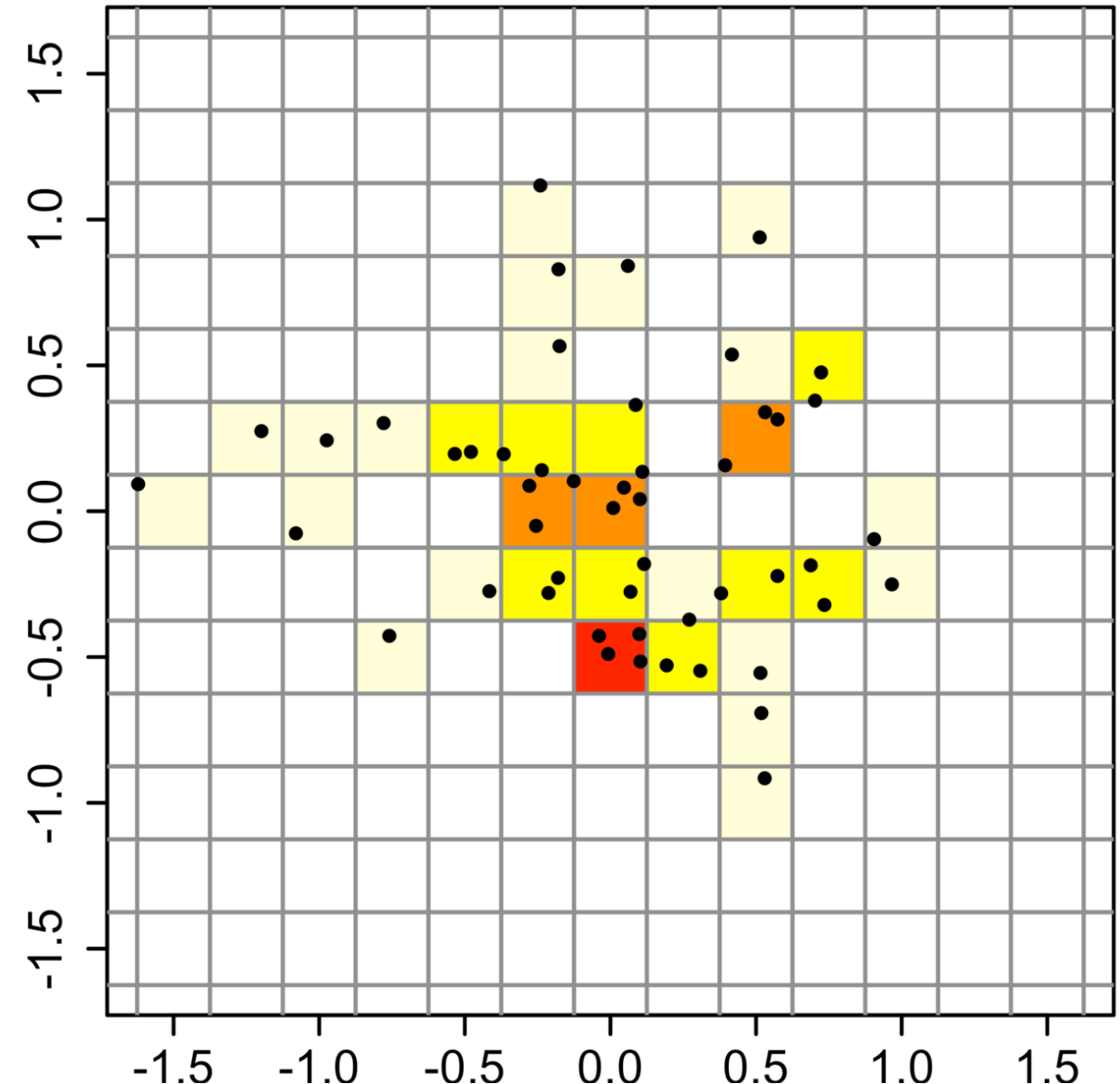
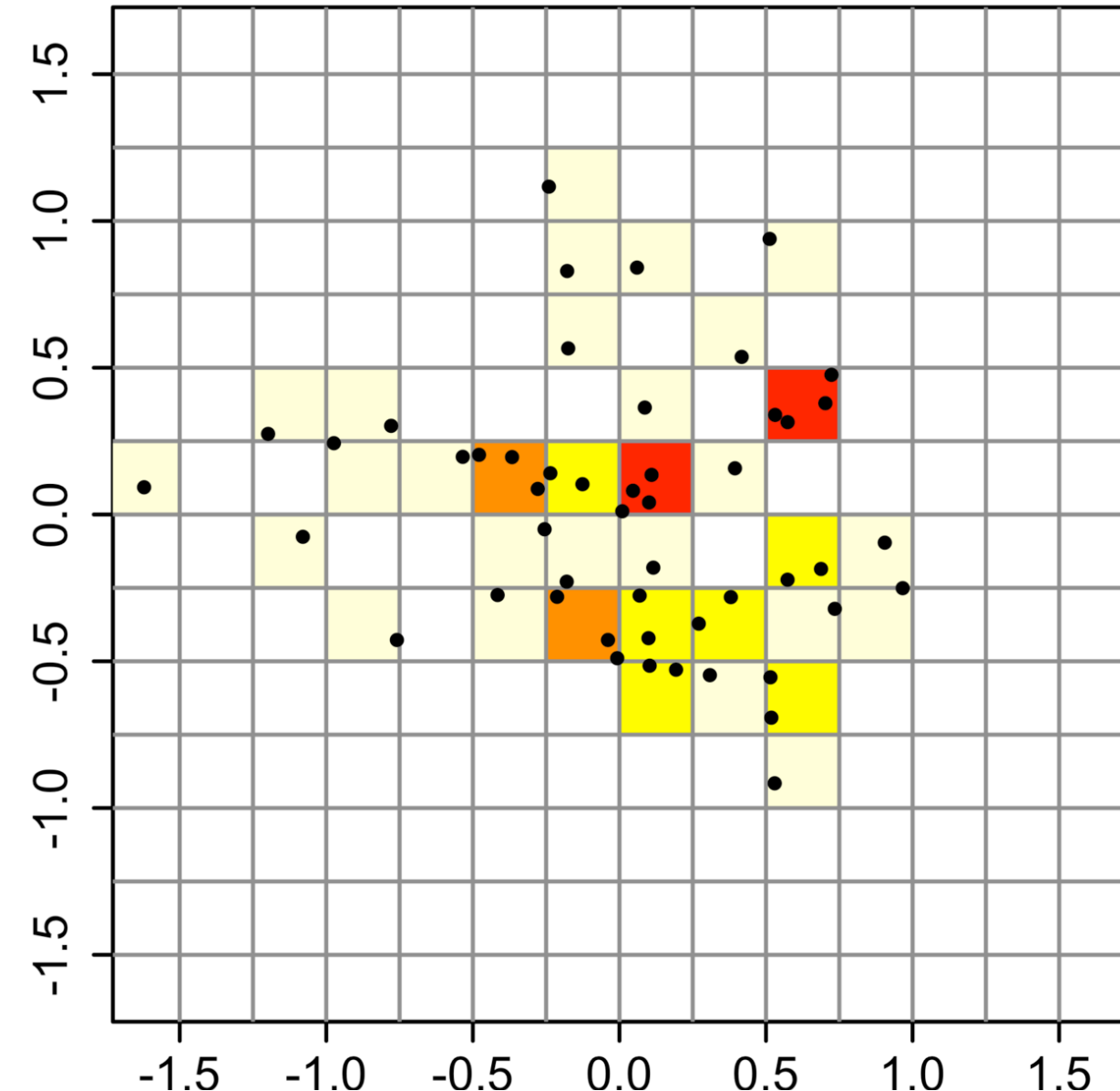
Nonparametric/Kernel Density Estimation

- Blue = true PDF
- Red = data set; i.i.d. drawn from PDF
- **Black** = kernel density estimate, for specific data set



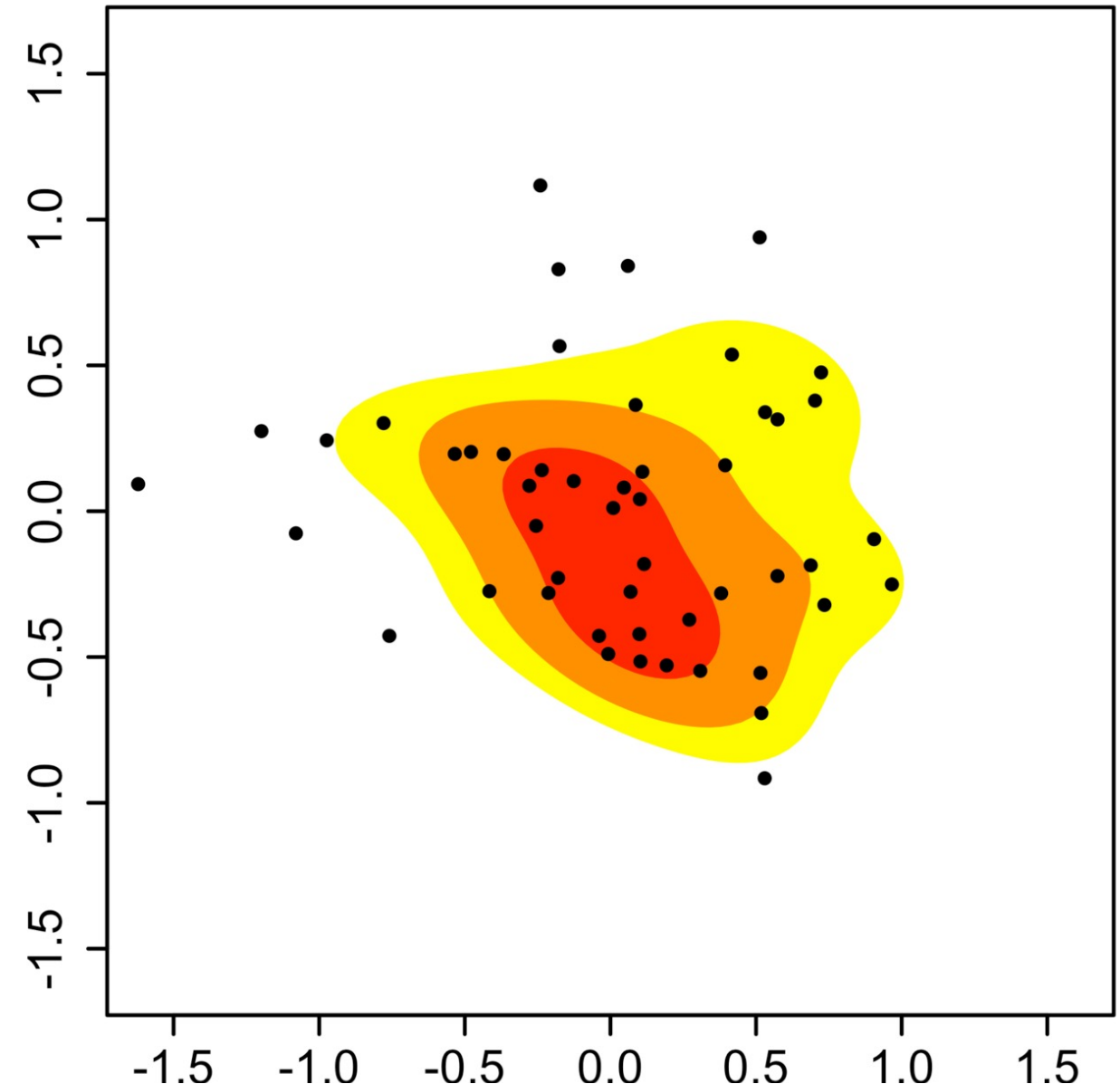
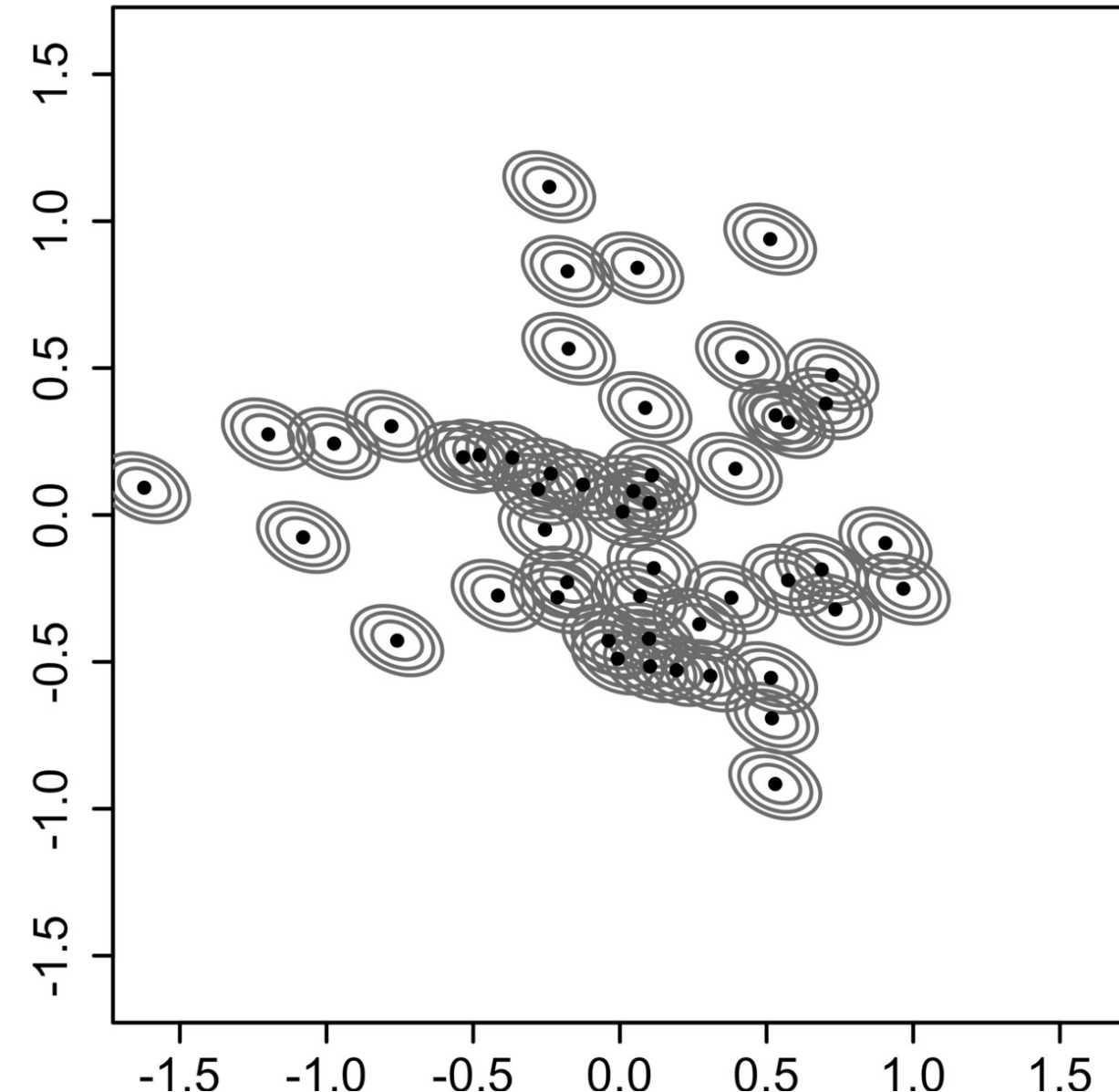
Nonparametric/Kernel Density Estimation

- Histogram (in 2D): depends on bin placement (and bin size)



Nonparametric/Kernel Density Estimation

- Kernel density estimate: depends on kernel (co)variance



Nonparametric/Kernel Density Estimation

- Goal: Estimate probability density function (PDF) $f(\mathbf{x})$ given observations $\{ \mathbf{x}_i \text{ in } \mathbb{R}^d : i = 1, \dots, n \}$ drawn from $f(\mathbf{x})$, where analytical/parametric form of $f(\mathbf{x})$ is unknown

- Strategy: Superpose kernel functions, each placed at $\mathbf{x}_i$

$$f(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^n K \left(\frac{\mathbf{x} - \mathbf{x}_i}{h} \right)$$

- h = scale/“bandwidth” parameter

- $K(y)$ = kernel

- e.g., standard normal $\exp(-0.5 \|\mathbf{y}\|_2^2) / (2\pi)^{d/2}$

- Non-negative, integrates to 1, mean 0, finite valued, decays to 0 sufficiently fast

- Typically, radially symmetric

- “Profile” (lowercase) $k(\cdot)$ = non-increasing;

For Gaussian kernel: profile $k(a) = \exp(-a/2)$ for $a > 0$

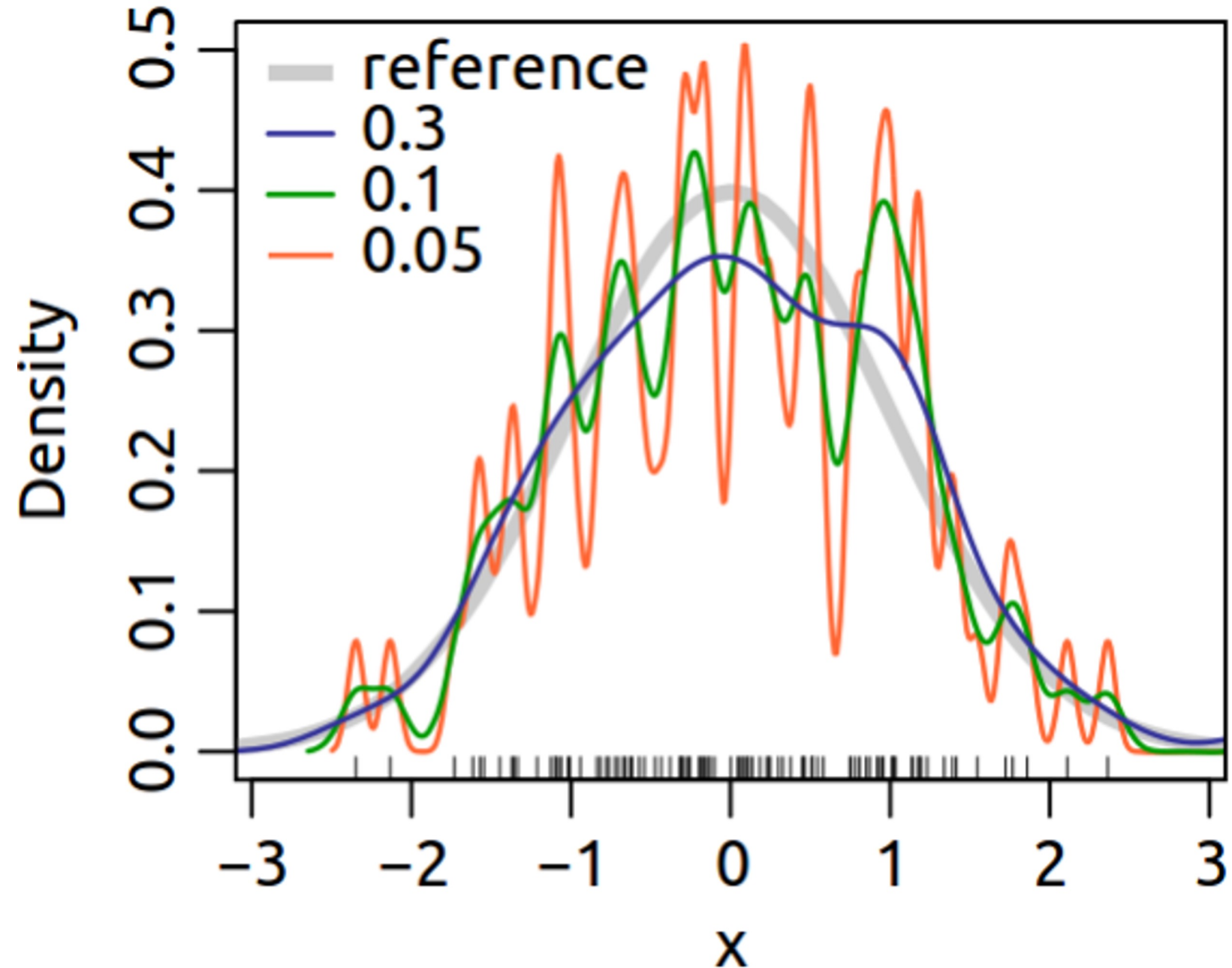
$$\int_{\mathbb{R}^d} K(\mathbf{x}) d\mathbf{x} = 1$$

$$\int_{\mathbb{R}^d} \mathbf{x} K(\mathbf{x}) d\mathbf{x} = \mathbf{0}$$

$$\lim_{\|\mathbf{x}\| \rightarrow \infty} \|\mathbf{x}\|^d K(\mathbf{x}) = 0$$

Nonparametric/Kernel Density Estimation

- Bandwidth/scale (h) selection
- **Too small: orange**
- **Small: green**
- **Desirable: blue**
- **True: gray**
- What if h is too large ?
- What if $h \rightarrow 0$?

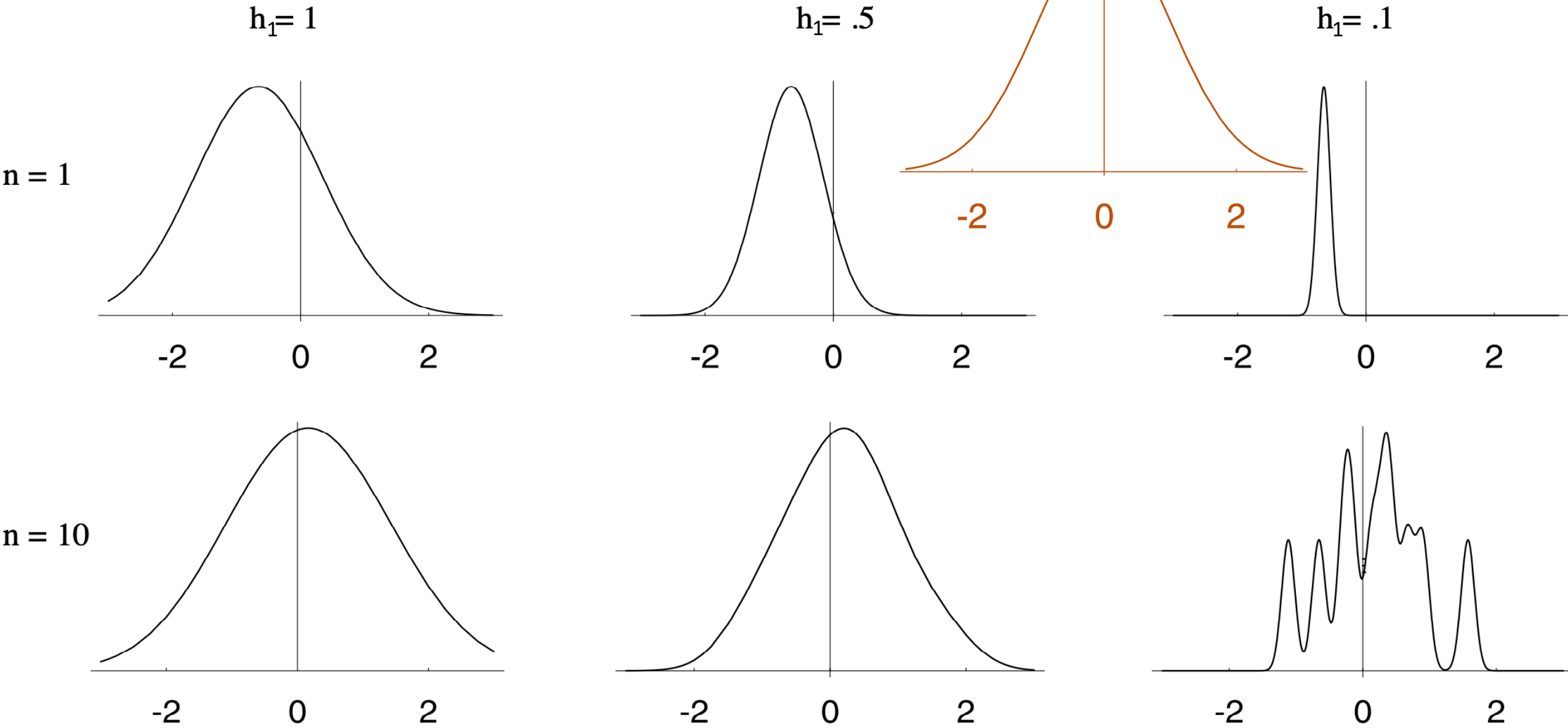


Nonparametric/Kernel Density Estimation

- How can PDF estimate converge to true PDF as sample size $n \rightarrow \infty$?
- Bandwidth constraints (**motivation**)
 - 1) h must reduce with n (**more data should need less smoothing**)
 - 2) $h(n) \rightarrow 0$ as $n \rightarrow \infty$ (**else convolution always gives blurred estimate**)
 - Sufficiently fast : $\lim_{n \rightarrow \infty} [h(n)]^d = 0$
 - Sufficiently slow: $\lim_{n \rightarrow \infty} n [h(n)]^d = \infty$ (decreases slower than $1/n$)
- Convergence
 - Let random variable $P_n(x; X_1, \dots, X_n)$ = kernel density estimate of PDF $P(\cdot)$ at x using a sample $\{X_1, \dots, X_n\}$ of cardinality n
 - Then, $\lim_{n \rightarrow \infty} E_{P(X_1, \dots, X_n)} [P_n(x; X_1, \dots, X_n)] = P(x)$
 - Expectation is w.r.t. sampling distribution, of n -sized sample drawn from $P(\cdot)$
 - And, $\lim_{n \rightarrow \infty} \text{Var} [P_n(x)] = 0$ (mean-square convergence)

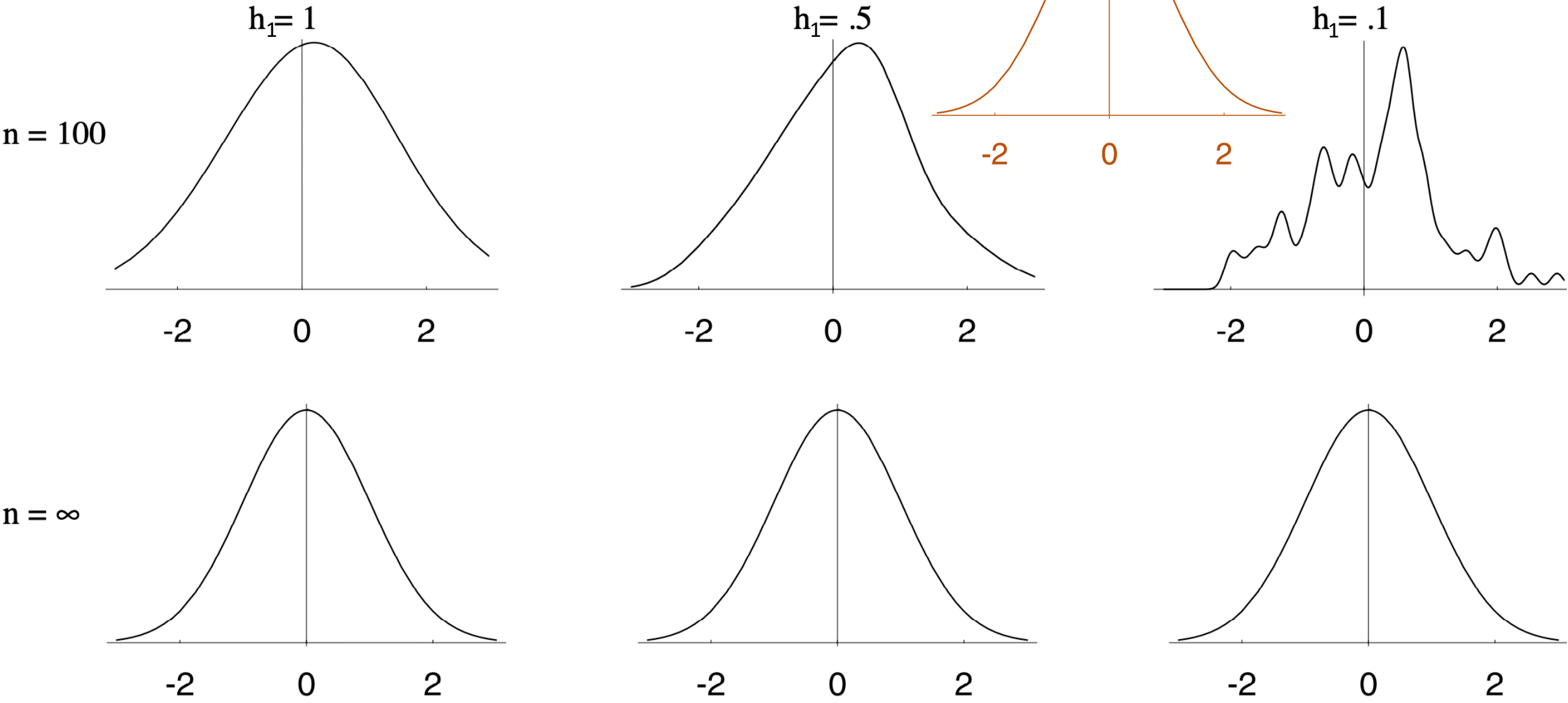
Nonparametric/Kernel Density Estimation

- Example 1: $h(n) := h_1 / \text{sqrt}(n)$; any $h_1 > 0$



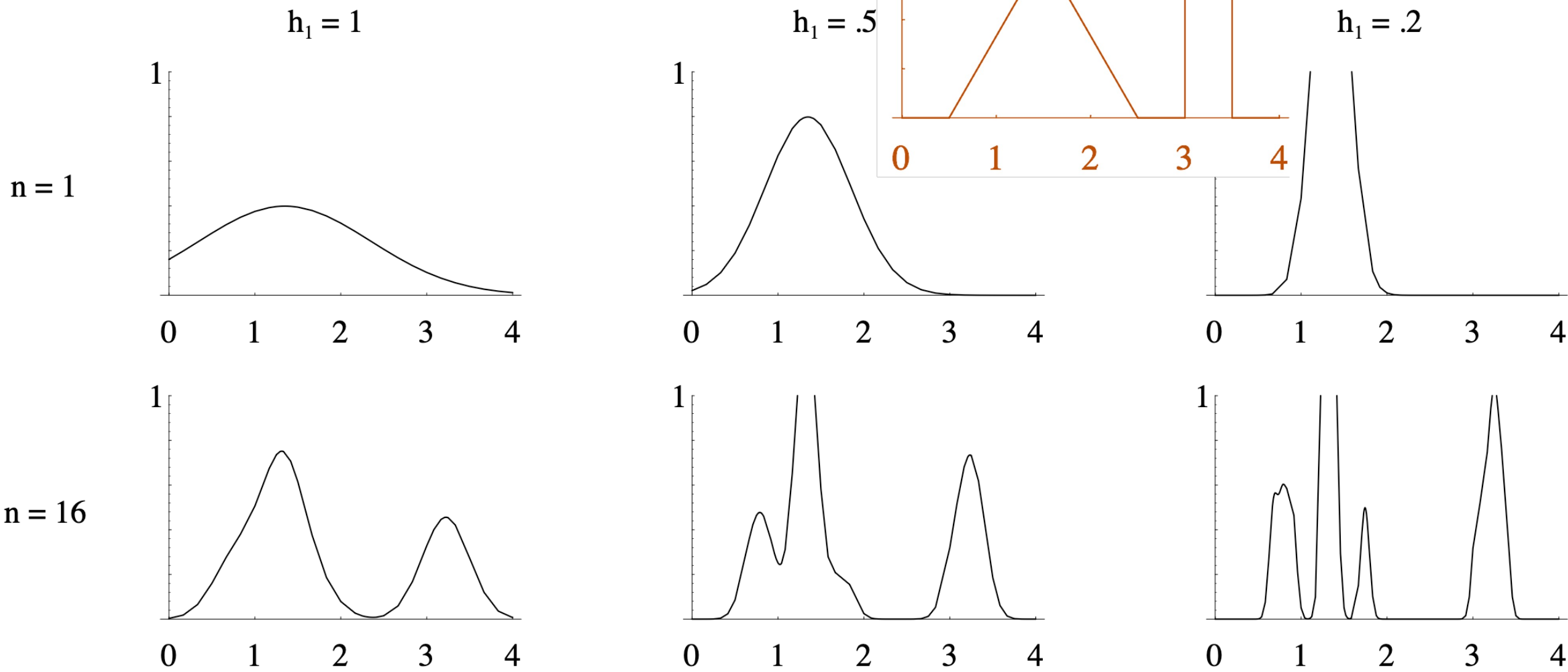
Nonparametric/Kernel Density Estimation

- Example 1: $h(n) := h_1 / \text{sqrt}(n)$; any $h_1 > 0$



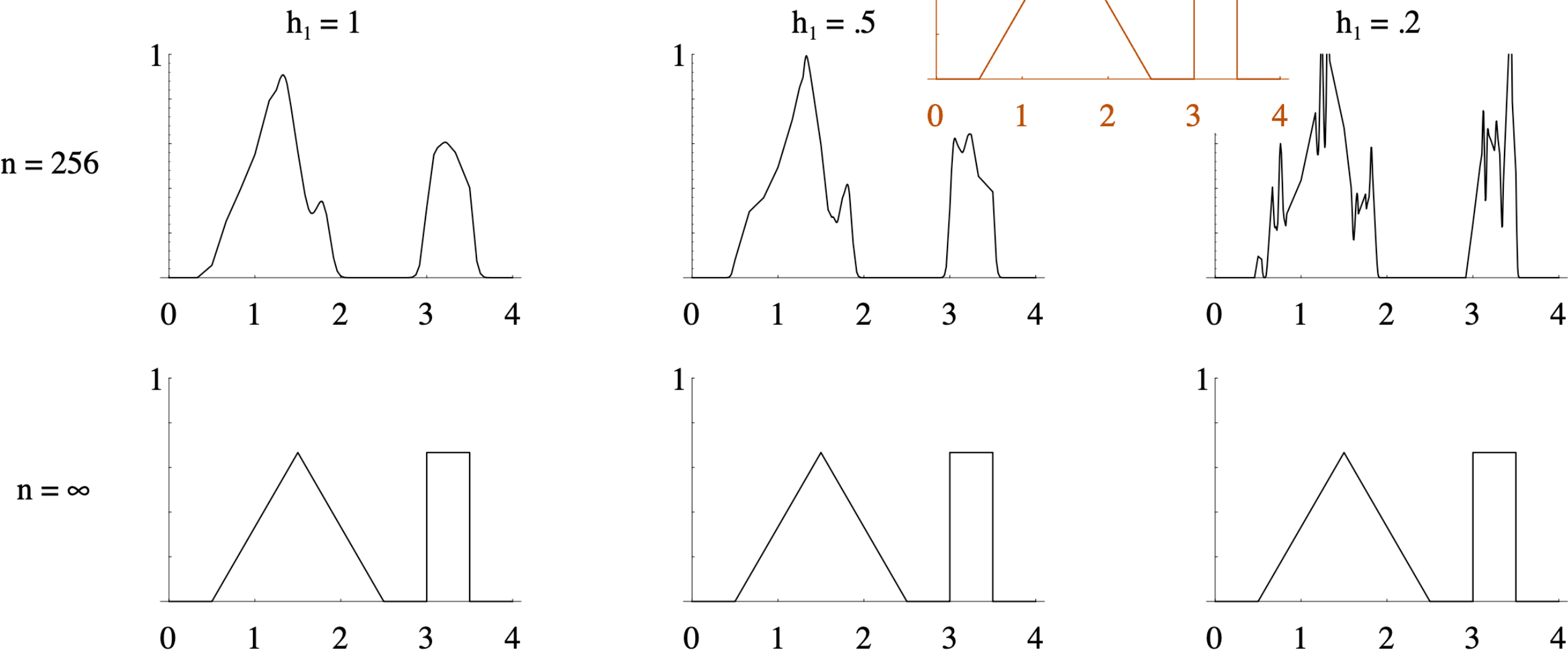
Nonparametric/Kernel Density Estimation

- Example 2: $h(n) := h_1 / \text{sqrt}(n)$
 - Specific choice of $h_1 > 0$ doesn't matter



Nonparametric/Kernel Density Estimation

- Example 2: $h(n) := h_1 / \text{sqrt}(n)$
 - Specific choice of $h_1 > 0$ doesn't matter



Nonparametric/Kernel Density Estimation

- Convergence of expectation/mean of PDF estimate
- Random variable $p_n(\mathbf{x}) := \text{PDF estimate at } \mathbf{x} \text{ with sample size } n$
 $:= (1/n) \sum_i (1/V_n) \varphi((\mathbf{x}-\mathbf{x}_i)/h_n)$ (e.g., $\varphi(\cdot)$ is standard normal)
where $V_n := (h_n)^d$
$$\bar{p}_n(\mathbf{x}) = \mathcal{E}[p_n(\mathbf{x})]$$
- Inside each expectation,
 $\mathbf{x}_i$ sampled from same $p(\mathbf{X})$
$$= \frac{1}{n} \sum_{i=1}^n \mathcal{E} \left[\frac{1}{V_n} \varphi \left(\frac{\mathbf{x} - \mathbf{x}_i}{h_n} \right) \right]$$
- So, all n expectations are same = $\bar{p}_n(\mathbf{x})$
- So, LHS = $\mathcal{E} \left[(1/V_n) \varphi((\mathbf{x}-\mathbf{x}_i)/h_n) \right]$
$$= \int \frac{1}{V_n} \varphi \left(\frac{\mathbf{x} - \mathbf{v}}{h_n} \right) p(\mathbf{v}) d\mathbf{v}$$
- = Conv (PDF $p(\cdot)$, kernel $\varphi(\cdot/h_n)/V_n$)
- If $V_n \rightarrow 0$, kernel $\rightarrow$ impulse function & convolution converges to $p(\mathbf{x})$;
this happens even for finite n (good; but what about variance ?)
 - If $V_n > 0$, then KD estimate is biased; for finite n , we prefer biased but smooth

Nonparametric/Kernel Density Estimation

- Variance of PDF estimate

- Random variable $p_n(\mathbf{x}) := \sum_i \varphi((\mathbf{x} - \mathbf{x}_i)/h_n) / (nV_n)$

= sum of functions of independent $\mathbf{x}_i$

$$\sigma_n^2(\mathbf{x}) = \sum_{i=1}^n \mathcal{E} \left[\left(\frac{1}{nV_n} \varphi \left(\frac{\mathbf{x} - \mathbf{x}_i}{h_n} \right) - \frac{1}{n} \bar{p}_n(\mathbf{x}) \right)^2 \right]$$

- So, $\text{Var}(p_n(\mathbf{x})) =$ sum of variances of its terms

$$= n \mathcal{E} \left[\frac{1}{n^2 V_n^2} \varphi^2 \left(\frac{\mathbf{x} - \mathbf{x}_i}{h_n} \right) \right] - \frac{1}{n} \bar{p}_n^2(\mathbf{x})$$

- Each term's variance is same

$$= \frac{1}{nV_n} \int \frac{1}{V_n} \varphi^2 \left(\frac{\mathbf{x} - \mathbf{v}}{h_n} \right) p(\mathbf{v}) d\mathbf{v} - \frac{1}{n} \bar{p}_n^2(\mathbf{x})$$

- Now $\varphi^2(.) \leq \varphi(.) \sup(\varphi(.))$

- So, integral $\leq \sup(\varphi(.)) E[p_n(\mathbf{x})]$

- So, $0 \leq \text{LHS} \leq \sup(\varphi(.)) E[p_n(\mathbf{x})] / (nV_n) - E[p_n(\mathbf{x})]^2 / n$

Nonparametric/Kernel Density Estimation

- Variance of PDF estimate
- We have: $0 \leq \text{Var}(p_n(x)) = \text{LHS} \leq \sup(\varphi(.)) E[p_n(x)] / (nV_n) - E[p_n(x)]^2 / n$
- As $n \rightarrow \infty$, earlier, we let $V_n \rightarrow 0$ to make mean of $p_n(x)$ converge to $p(x)$
- So, as $n \rightarrow \infty$ & $V_n \rightarrow 0$, second term $\rightarrow 0$ (assume $E[p_n(x)] = p(x)$ finite)
- Also, as $n \rightarrow \infty$, we want first term $\rightarrow 0$ to achieve zero variance of $p_n(x)$ for which we can let $nV_n \rightarrow \infty$
 - $\sup(\varphi(.))$ is finite, e.g., when $\varphi(.)$ is standard normal
- Typical choices for V_n are $V_n := V_1 / \log(n)$ or $V_n := V_1 / \sqrt{n}$

Nonparametric/Kernel Density Estim

- Books
- Notes: [link](#)

